

# GRAPH

## PROBLEM 4

### QUESTION:

| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Session | Graph | Question Information | Level 1 | Challenge 84 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|----------------------|---------|--------------|
| <p><b>Question description</b></p> <p>Byteland has <math>n</math> cities and <math>m</math> roads between them. The goal is to construct new roads so that there is a route between any two cities. Your task is to find out the minimum number of roads required, and also determine which roads should be built.</p> <p><b>Constraints</b></p> $1 \leq n \leq 10^5$ $1 \leq m \leq 2 \cdot 10^5$ $1 \leq a, b \leq n$ <p><b>Input</b></p> <p>The first input line has two integers <math>n</math> and <math>m</math>: the number of cities and roads. The cities are numbered <math>1, 2, \dots, n</math>.</p> <p>After that, there are <math>m</math> lines describing the roads. Each line has two integers <math>a</math> and <math>b</math>: there is a road between those cities.</p> <p>A road always connects two different cities, and there is at most one road between any two cities.</p> <p><b>Output</b></p> <p>First print an integer <math>k</math>: the number of required roads.</p> <p>Then, print <math>k</math> lines that describe the new roads. You can print any valid solution.</p> |         |       |                      |         |              |

#### Logical Test Cases

| Test Case 1                                                                   | Test Case 2                                                                   |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| INPUT (STDIN)<br><br>4 2<br>1 2<br>3 4<br><br>EXPECTED OUTPUT<br><br>1<br>2 4 | INPUT (STDIN)<br><br>4 2<br>2 3<br>2 4<br><br>EXPECTED OUTPUT<br><br>1<br>1 4 |

#### Mandatory Test Cases

| Test Case 1               |
|---------------------------|
| KEYWORD<br><br>while(m--) |

#### Complexity Test Cases

| Test Case 1                    | Test Case 2            | Test Case 3    |
|--------------------------------|------------------------|----------------|
| CYCLOMATIC COMPLEXITY<br><br>5 | TOKEN COUNT<br><br>475 | NLOC<br><br>50 |

## SOURCE CODE:

```
#include <stdio.h>

int parent_city[100005];

int resolve_leader(int city) {
    if (parent_city[city] == city) return city;
    return parent_city[city] = resolve_leader(parent_city[city]);
}

void merge_networks(int a, int b) {
    int ra = resolve_leader(a);
    int rb = resolve_leader(b);
    if (ra != rb) parent_city[ra] = rb;
}

int main() {
    int total_cities, roads;
    if (scanf("%d %d", &total_cities, &roads) != 2) return 0;

    for (int c = 1; c <= total_cities; c++) parent_city[c] = c;

    for (int i = 0; i < roads; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        merge_networks(u, v);
    }

    int hubs[100005];
    int hub_count = 0;
    for (int c = 1; c <= total_cities; c++) {
        if (resolve_leader(c) == c) {
            hubs[hub_count++] = c;
        }
    }

    int new_roads_needed = hub_count - 1;
    printf("%d\n", new_roads_needed);
    for (int i = 0; i < new_roads_needed; i++) {
        printf("%d %d\n", hubs[i], hubs[i + 1]);
    }
    return 0;
}
```

## OUTPUT:

```
4 2
1 2
3 4
1
2 4
```

```
4 2
2 3
2 4
1
1 4
```